AI Semiconductor Value Chain

Fundamental Screening, Blended Valuation, and a Black–Litterman Portfolio Across the End-to-End Stack

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Pipeline run with SEC EDGAR, Yahoo Finance and FRED data through September 2, 2026

Abstract

This paper documents a reproducible market-intelligence pipeline that screens 116 US-listed companies across eleven segments of the AI and semiconductor value chain, ranks them on a four-pillar composite (value, growth, quality, estimate momentum), values each name with a through-cycle DCF, a cross-sectional multiple regression and consensus targets, and converts the resulting top-15 list into a constrained Black–Litterman portfolio. The September 2026 run places Sandisk, Micron and NVIDIA at the top of the ranking, followed by Credo, Astera Labs and Nebius; the maximum-Sharpe allocation concentrates in memory, NVIDIA and two equipment sub-suppliers while holding growth names at the 2% floor. The principal conclusion is that the growth-adjusted valuation dispersion inside the chain is now larger than the dispersion between the chain and the market: consensus already prices extraordinary growth for interconnect and neocloud names, while memory and manufacturing enablers trade at a fraction of the multiple their fundamentals would justify. Because the backtest reuses today's information, the allocation is a forward policy portfolio and not evidence of realised performance.

Research-use notice. This document is an analytical template, not investment advice. Section 11 reports the point-in-time validation that superseded the scoring weights of Section 4. Company fields are drawn from SEC XBRL company facts and Yahoo Finance consensus and have been cross-checked but not audited; PitchBook was not connected for this run. Fair values are model outputs under the assumptions stated in Appendix B.

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1 Investment conclusion

The research supports a **bottlenecks-with-valuation-discipline** portfolio rather than a basket of the best-known AI names. The 15 selected securities perform different economic jobs, carry very different multiples, and—critically—different amounts of growth already embedded in their prices.

- **Memory supercycle—SNDK, MU, STX.** High-bandwidth memory, NAND and nearline storage carry the strongest combination of consensus growth (50–120% next twelve months), 90-day estimate revisions (+33% to +40%) and single-digit forward earnings multiples. They lead the ranking but are the most cyclical exposure in the list.
- **Compute and interconnect—NVDA, AVGO, CRDO, ALAB, AAOI.** Accelerators, custom silicon and the electrical/optical links between them. NVIDIA and Broadcom combine scale with 43–47% free-cash-flow margins; the three interconnect names grow faster but trade at multiples that already discount post-consensus growth above 45%.
- **Manufacturing and infrastructure enablers—KLIC, UCTT, CLS, DELL, DIOD.** Advanced-packaging bonders, gas-delivery subsystems, hyperscale hardware manufacturing, AI servers and discretes. These names trade at 1.0–2.5× next-twelve-month sales against regression-fair multiples of 2.3–5.6× and receive the largest optimiser weights.
- **AI monetisation and capacity—NBIS, APP.** Neocloud GPU capacity and an AI-driven advertising engine. Both are valuation-sensitive: Nebius has negative free cash flow and only two usable valuation methods; AppLovin's composite rests on quality and value pillars after a 34% twelve-month decline.

The composite score is strongest for SNDK (1.27), MU (1.20), NVDA (0.99) and CRDO (0.89), followed by ALAB (0.68) and NBIS (0.64). Those rankings prioritise research effort. They are not position weights: covariance, valuation confidence, beta and liquidity are incorporated separately in Section 6.

1.1 Key findings

1. **Growth is extreme and dispersed.** Consensus next-twelve-month revenue growth ranges from 23% for Diodes to 347% for Nebius, and the universe median is 26%. Winsorised z-scores and a per-segment cap are required so that memory and neocloud growth does not monopolise the list.
2. **The market already prices the growth of the fastest growers.** The reverse DCF shows that CRDO, ALAB, AAOI and NBIS need post-consensus growth of 46% or more (at today's margins) to justify their prices, whereas SNDK, MU and DIOD are priced for 0–23%. The composite rewards the second group through its value pillar.
3. **Regression multiples and DCF disagree by design.** Across the universe the median DCF upside is −52% while the median regression-multiple upside is −3%: a 4.75% ten-year yield and 10–11% discount rates are unforgiving for long-duration cash flows. The blend (50/30/20) and the 0.1–5× sanity filter keep either method from dominating.

4. **Optimiser weights are corner solutions.** With 3-year annualised volatilities of 47–129% and average pairwise correlation of 0.43, the maximum-Sharpe portfolio pins eight names at the 2% floor and three at the 15% cap. Expected return is 14.2% at 46.5% volatility (Sharpe 0.20); the equal-weight basket is dominated.

5. **Daily monitoring is part of the method.** The pipeline re-scores the universe every trading day, re-optimises, and raised 41 alerts on this run—including a −20% session move in CRDO and two names (SMCI, CRWV) leaving the list after the valuation recalibration described in Section 5.

2 Universe and value-chain map

The universe deliberately spans the physical and economic flow of the AI stack: design tools, wafer-fabrication equipment, foundry and packaging, accelerators, memory, interconnect, analog and power silicon, servers and data-centre infrastructure, cloud and neocloud capacity, AI software, and power generation for data centres. Table 1 summarises the segments; nine US-listed ADRs of foreign issuers (TSM, ASML, ARM, NXPI, UMC, NBIS, CAMT, NVMI, IREN) are included and flagged.

Table 1: Universe by value-chain segment (segment medians unless stated)

Segment	N	Mkt cap	NTM gr.	GM	EV/S	P/E	Upside	Structure (HHI)
EDA & semiconductor IP	3	$412B	15%	86%	12.4x	38.1x	−26%	Concentrated (3,514)
Semiconductor equipment	20	$1.74T	29%	47%	4.9x	25.4x	−25%	Moderate (1,889)
Foundry, IDM & packaging	5	$2.70T	15%	31%	4.6x	20.4x	−22%	Concentrated (4,784)
AI compute silicon	5	$8.32T	64%	56%	9.8x	20.1x	−6%	Concentrated (4,117)
Memory & storage	5	$1.67T	50%	71%	4.6x	19.6x	+10%	Concentrated (4,797)
Interconnect, networking & optics	13	$991B	39%	58%	8.8x	30.4x	−27%	Concentrated (4,214)
Analog, power & mixed-signal	18	$678B	19%	53%	6.1x	26.7x	−15%	Fragmented (1,385)
Servers & DC infrastructure	12	$877B	25%	17%	1.6x	18.5x	+11%	Moderate (1,745)
Cloud, hyperscale & neocloud	12	$13.0T	23%	63%	5.7x	21.6x	+32%	Concentrated (2,529)
AI software & applications	15	$1.25T	20%	77%	8.0x	41.0x	−6%	Moderate (2,138)
Power & energy for AI	8	$501B	26%	28%	4.6x	17.9x	−24%	Moderate (2,404)

Source: Pipeline output `segment_table.csv`. Market cap is the segment aggregate; EV/S and P/E are next-twelve-month (NTM); upside is the segment median blended fair-value upside; HHI is computed on trailing revenue within the segment.

Table 2: Selected 15 by economic role

Lane	Constituents	Names	Avg. score	Med. NTM growth	Med. EV/S
Memory supercycle	SNDK, MU, STX	3	0.96	86%	4.4x
Compute & interconnect	NVDA, AVGO, CRDO, ALAB, AAOI	5	0.71	76%	9.9x
Manufacturing & infra. enablers	KLIC, UCTT, CLS, DELL, DIOD	5	0.47	36%	1.5x
AI monetisation & capacity	NBIS, APP	2	0.55	191%	8.5x

Source: Author grouping of the pipeline's top 15; medians are of the constituents' NTM consensus revenue growth and EV/NTM sales.

The economic thesis is that AI spending migrates through a chain of bottlenecks. Accelerator demand creates HBM, packaging and interconnect demand; denser racks require power conversion, cooling and hyperscale manufacturing capacity; deployed capacity must eventually be monetised through cloud and software. The industry map (Figure 1) shows the selection concentrating in

memory, compute and interconnect—the three most concentrated segments—while the enablers lane provides cheaper exposure to the same capital-spending shock.

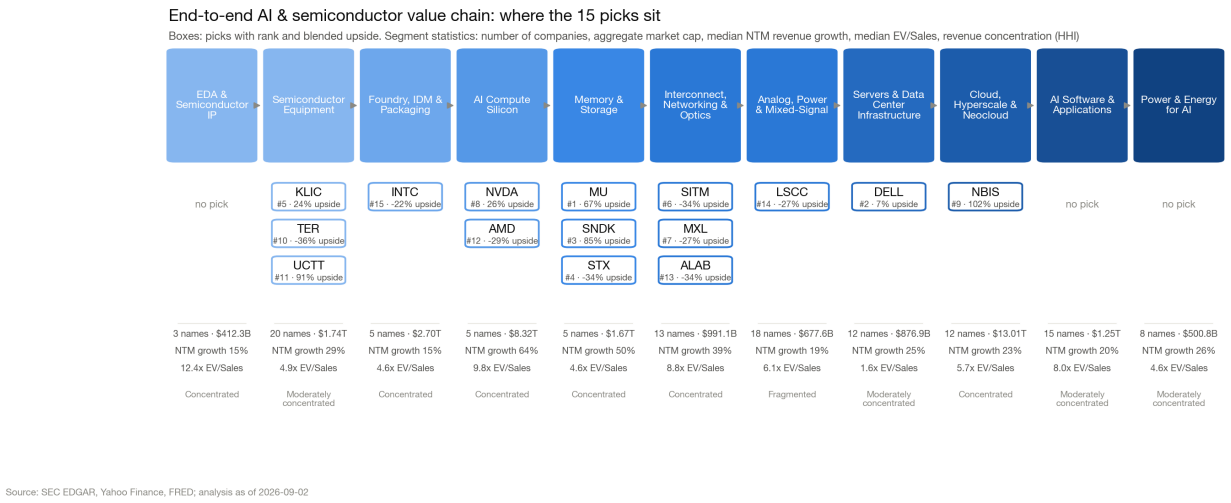


Figure 1: End-to-end value chain with the position of the 15 picks and segment statistics

3 Data and evidence

3.1 Data sources

Three free, official or quasi-official sources feed the model; a fourth is optional.

- **SEC EDGAR.** XBRL company facts (revenue, gross profit, R&D, operating income, operating cash flow, capital expenditure, share-based compensation, buybacks, equity, cash, debt, diluted EPS) and the filing index. Annual series are de-duplicated by fiscal-year end with the latest restatement winning; 20-F filers are read through the same aliases where IFRS tags exist. The filing index drives the "new 10-Q / 8-K" alert. 116 of 116 names resolved to a CIK.
- **Yahoo Finance (via `yfinance`).** Four years of adjusted prices for the universe and three benchmarks (SPY, QQQ, SOXX), annual and quarterly statements, consensus revenue and EPS for the current and next fiscal year, the 7/30/60/90-day EPS trend, price targets, the earnings calendar and spot FX.
- **FRED.** Sixteen series including the ten-year Treasury yield (the risk-free rate), the two-year yield, fed funds, CPI and core PCE, unemployment, real GDP, industrial production, semiconductor industrial production (IPG3344S) and producer prices (PCU33443344), the VIX, high-yield spreads, the dollar and core capital-goods orders.
- **PitchBook.** An adapter ingests platform exports or a licensed API when credentials exist. Neither was available for this run; the paper therefore contains no private-market comparables.

Two adjustments matter for cross-sectional comparability. First, statements and consensus of foreign filers are reported in their home currency (TSM in TWD, ASML in EUR); the pipeline

converts them at spot FX, implies the ADR-equivalent share count from the USD market capitalisation, and recomputes enterprise value as market capitalisation plus net debt rather than trusting provider EV fields that mix currencies. Second, Yahoo's "current year" estimate refers to the first unreported fiscal year; next-twelve-month (NTM) figures blend the current and next fiscal year by the months remaining, which aligns Micron (August year-end) with Sandisk (June) and NVIDIA (January).

3.2 Macro backdrop

Table 3: Macro snapshot from FRED

Series	Indicator	Latest	3m chg	12m chg
DGS10	10-year Treasury yield	4.75%	+0.30	+0.52
DGS2	2-year Treasury yield	4.34%	+0.36	+0.75
FEDFUNDS	Fed funds effective rate	3.63%	0.00	−0.70
CPI / core PCE	Inflation, YoY	3.5% / 3.3%	−0.41 / +0.01	+0.80 / +0.48
UNRATE	Unemployment rate	4.1%	−0.20	−0.20
GDPC1	Real GDP, YoY	2.1%	−0.59	+0.02
IPG3344S	Semiconductor industrial production, YoY	+11.9%	+3.1	−7.8
PCU33443344	Semiconductor producer prices, YoY	+27.1%	+1.6	+26.9
NEWORDER	Core capital-goods orders, YoY	+12.6%	+1.6	+8.8
VIXCLS / HY OAS	Volatility and credit	16.3 / 2.65%	+0.29 / −0.07	+0.22 / −0.19

Source: FRED, observations through August 2026 (monthly series through July 2026). Changes are in percentage points except VIX (index points).

The backdrop is a still-expanding semiconductor cycle—output up 11.9% and producer prices up 27.1% year on year, with core capital-goods orders accelerating—against a higher-for-longer discount rate. The ten-year yield is 52 basis points above a year ago even though the policy rate has been cut by 70 basis points, so the curve has re-steepened. For long-duration AI cash flows this is the single most important valuation input, and it is why the DCF is the most conservative of the three methods throughout this paper.

3.3 Price evidence

Table 4 summarises the trailing price behaviour of the selection. Three facts stand out. Memory has already re-rated violently (Sandisk up 29×, Micron 8× over twelve months); the composite still ranks them first because consensus estimates have risen even faster than prices. Second, ten of the fifteen names drew down 38–66% within the last year, so intra-year path risk is severe even for winners. Third, six-month relative strength versus SOXX is negative for eight names: the value and quality pillars, not momentum, put them on the list.

4 Feature engineering and scoring architecture

4.1 Feature set

For each company i the pipeline computes roughly sixty fields from the raw statements, consensus and prices. Table 5 lists the ones that enter the score.

Table 4: Trailing price statistics for the selection

Ticker	Lane	12m return	1y max DD	1y vol	Beta (SPY)	6m vs SOXX
SNDK	Memory	+2,912%	−57%	117%	2.41	+110 pts
MU	Memory	+698%	−39%	81%	2.34	+89
NVDA	Compute	+32%	−20%	38%	2.05	−23
CRDO	Interconnect	+34%	−54%	96%	2.76	+23
ALAB	Interconnect	+55%	−60%	98%	2.72	+91
NBIS	Capacity	+204%	−48%	119%	2.61	+58
KLIC	Enabler	+114%	−42%	57%	1.72	−27
CLS	Enabler	+45%	−40%	77%	2.44	−37
UCTT	Enabler	+174%	−54%	87%	2.52	−37
AVGO	Compute	+25%	−29%	48%	2.02	−30
AAOI	Interconnect	+333%	−66%	146%	3.75	−46
APP	Monetisation	−34%	−59%	74%	2.26	−75
STX	Memory	+380%	−32%	74%	1.56	+77
DIOD	Enabler	+63%	−38%	67%	1.89	−14
DELL	Enabler	+284%	−32%	72%	1.89	+166
SOXX	Benchmark	+108%	−41%	40%	—	—

Source: Yahoo Finance adjusted closes through September 2, 2026. Beta is realised against SPY over three years of daily returns (recent IPOs proxied with segment peers before listing). SOXX drawdown is over the three-year backtest window.

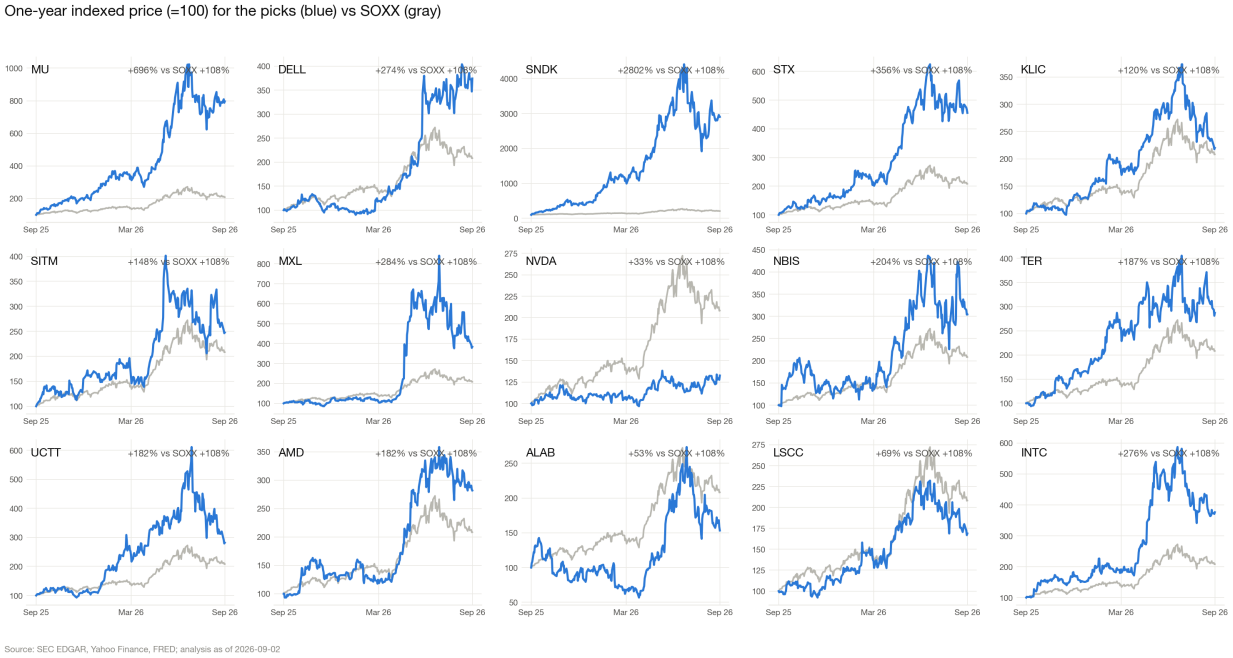


Figure 2: One-year indexed price for each pick (blue) against SOXX (grey)

Table 5: Scoring features by pillar

Pillar (weight)	Inputs (direction)	Treatment
Value (0.40)	EV/S_{NTM} to growth (−), P/E_{NTM} (−), PEG (−), FCF yield (+), earnings yield (+), DCF upside (+), regression residual (−), Street-target upside (+)	Half universe-wide z, half segment-relative z, so a 1× EMS name is not automatically "cheaper" than a 12× software name
Growth (0.30)	NTM and FY+1 revenue growth, 3-year revenue CAGR, latest-quarter YoY, 2-year forward CAGR, NTM EPS growth (all +)	Universe-wide z
Quality (0.20)	Gross, operating and FCF margins, ROIC, Rule of 40, 3-year operating-margin change (+); net debt/EBITDA, SBC/revenue (−)	Universe-wide z
Momentum (0.10)	30- and 90-day consensus EPS revisions, 6-month return relative to SOXX, distance to 200-day average (all +)	Universe-wide z

4.2 Robust normalisation

Each raw metric $x_{i,k}$ is winsorised at the 5th and 95th percentiles of the eligible universe and standardised,

$$z_{i,k} = d_k \, \frac{\tilde{x}_{i,k} - \overline{\tilde{x}}_k}{\sigma(\tilde{x}_k)}, \qquad d_k \in \{-1, +1\}, \tag{1}$$

where $\tilde{x}$ denotes the winsorised value and d_k orients every field so that higher is better. Pillar scores are the mean of available z's, clipped to $[-3, 3]$; missing fields are neutral rather than favourable, and a name must have at least 60% feature coverage to be eligible. The composite is

$$S_i = 0.40\, V_i + 0.30\, G_i + 0.20\, Q_i + 0.10\, M_i. \tag{2}$$

4.3 Selection

Eligibility requires market capitalisation of at least $2 billion, 60% feature coverage and a computable blended fair value. Of 116 names, 112 pass those gates and 52 score above the universe mean. The top 15 are chosen greedily by composite with at most three names per segment, which forces the list to span at least five layers of the chain. Each pick is tagged *Undervalued*, *High-Growth* or *GARP* according to whether its value or growth pillar dominates by more than 0.25.

5 Valuation

5.1 Discounted cash flow

The DCF is a ten-year free-cash-flow-to-firm model. Year-1 revenue is NTM consensus. Growth beyond the consensus horizon starts at 60% of the FY+1 consensus rate, capped at 30%, and fades linearly to the 3% terminal rate by year 8:

$$g_1 = \min\!\big(0.6\, g_{\text{cons}}^{\text{FY+1}},\, 0.30\big), \qquad g_t = g_1 + (g_T - g_1)\,\frac{t-1}{7} \text{ for } t \le 8, \qquad g_t = g_T = 0.03 \text{ for } t > 8. \tag{3}$$

Table 6: September 2026 ranked selection with pillar scores

Rank	Ticker	Segment	Style	S	V	G	Q	M
1	SNDK	Memory & storage	High-Growth	1.27	0.88	1.74	1.29	1.39
2	MU	Memory & storage	High-Growth	1.20	0.95	1.65	0.95	1.41
3	NVDA	AI compute silicon	High-Growth	0.99	0.74	1.45	1.21	0.18
4	CRDO	Interconnect & optics	High-Growth	0.89	0.67	1.55	0.83	−0.10
5	ALAB	Interconnect & optics	High-Growth	0.68	−0.38	1.51	0.89	1.96
6	NBIS	Cloud & neocloud	High-Growth	0.64	−0.27	2.29	−0.44	1.54
7	KLIC	Semiconductor equipment	GARP	0.59	0.79	0.70	0.06	0.53
8	CLS	Servers & DC infrastructure	GARP	0.57	0.74	0.82	0.16	−0.06
9	UCTT	Semiconductor equipment	Undervalued	0.51	1.15	0.18	−0.51	0.96
10	AVGO	AI compute silicon	GARP	0.50	0.52	0.75	0.59	−0.49
11	AAOI	Interconnect & optics	High-Growth	0.48	−0.01	2.27	−0.60	−0.85
12	APP	AI software	Undervalued	0.46	0.75	0.02	1.40	−1.24
13	STX	Memory & storage	High-Growth	0.40	−0.18	0.41	0.81	1.89
14	DIOD	Analog & power	Undervalued	0.35	0.99	−0.24	−0.19	0.64
15	DELL	Servers & DC infrastructure	Undervalued	0.35	0.34	0.05	0.00	1.99

Source: Pipeline output `top15.csv`. Next in line, in order: SMCI (0.34), CRWV (0.32), COHU (0.31), BE (0.29), AMD (0.28).

Table 7: Fundamental profile of the selection

Ticker	NTM gr.	FY+1 gr.	3y CAGR	GM	OpM	FCF mgn	ROIC	ND/EBITDA	EPS rev 90d
SNDK	120%	18%	49%	72%	79%	34%	33%	−0.3x	+33%
MU	86%	86%	7%	73%	80%	29%	49%	−0.3x	+38%
NVDA	75%	63%	100%	75%	66%	47%	79%	−0.3x	+13%
CRDO	76%	55%	94%	68%	36%	31%	18%	−3.0x	+8%
ALAB	78%	56%	120%	75%	23%	23%	14%	−4.4x	+43%
NBIS	347%	258%	240%	74%	0%	−361%	5%	0.8x	+97%
KLIC	35%	32%	−24%	53%	21%	4%	14%	−2.5x	+27%
CLS	69%	71%	20%	12%	10%	3%	36%	0.2x	+20%
UCTT	39%	42%	−5%	16%	5%	−5%	4%	3.7x	+47%
AVGO	64%	64%	24%	76%	49%	43%	19%	1.1x	+4%
AAOI	147%	157%	27%	29%	−13%	−101%	−3%	n/m	−14%
APP	34%	27%	25%	89%	78%	66%	69%	0.1x	−3%
STX	50%	34%	18%	46%	43%	26%	59%	0.5x	+40%
DIOD	23%	21%	−10%	32%	8%	9%	5%	−1.6x	+31%
DELL	36%	12%	4%	19%	9%	7%	34%	1.4x	+34%

Source: SEC EDGAR company facts and Yahoo Finance consensus. Margins are trailing twelve months; ROIC is after-tax EBIT over invested capital; negative net debt denotes net cash. NBIS's FCF margin reflects capex several times revenue.

The FCF margin converges over five years from today's level m_0 to a through-cycle target that blends the current margin with the five-year average from EDGAR annual facts,

$$m^{*} = w\, m_0 + (1 - w)\, \bar{m}_{5y}, \qquad w = \begin{cases} 0.85 & \text{if 3-year revenue CAGR} > 25\%, \\ 0.60 & \text{otherwise,} \end{cases} \tag{4}$$

so that cyclicals such as Micron (five-year average -6%, target 15%) mean-revert while companies that have structurally scaled (Credo, target 25%) are not penalised for their sub-scale history. The discount rate is a CAPM cost of equity with a Blume-adjusted beta $\beta^{*} = 0.67\beta + 0.33$ bounded to $[0.7, 1.4]$, an equity risk premium of 4.5% over the ten-year yield of 4.75%, blended with after-tax debt at market weights and bounded to $[7.5\%, 11\%]$:

$$\mathrm{EV} = \sum_{t=1}^{10} \frac{R_t\, m_t}{(1 + \mathrm{WACC})^{t}} + \frac{R_{10}\, m_{10}\, (1 + g_T)}{(\mathrm{WACC} - g_T)(1 + \mathrm{WACC})^{10}}, \qquad \mathrm{FV} = \frac{\mathrm{EV} - \text{net debt}}{\text{shares}}. \tag{5}$$

Most selected names sit at the 11% cap: their raw betas are 1.4–3.8.

5.2 Reverse DCF

The market-implied growth g^{impl} is the post-consensus rate (same fade, margins held at today's level) that equates the DCF value to the share price. It converts a multiple into a statement about expectations that can be compared directly with consensus and with history. The search is bounded at 80%; names that hit the bound are either priced for hypergrowth or have current margins too thin to support the price at any growth rate.

5.3 Cross-sectional multiple regression

Across the 115 names with valid inputs, the log of the NTM enterprise-value-to-sales multiple is regressed on consensus growth and margins:

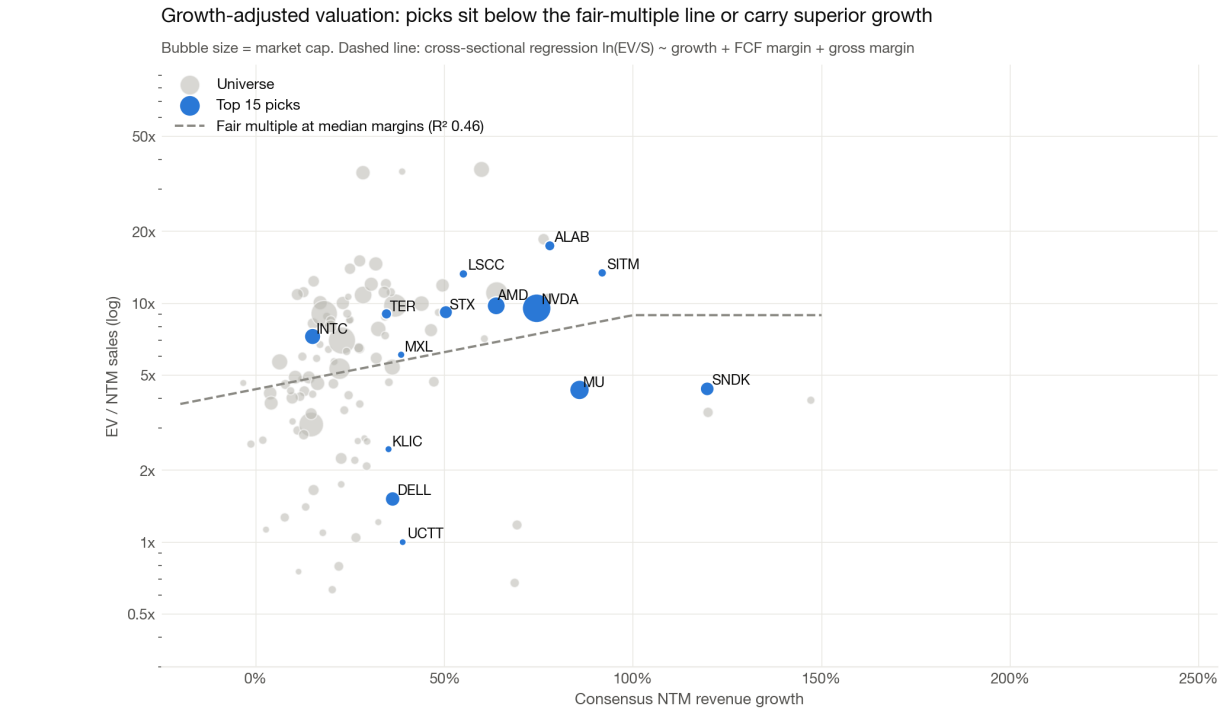
$$\ln(\mathrm{EV}/\mathrm{S}_{\mathrm{NTM}}) = 0.184 + 0.714\, g_{\mathrm{NTM}} + 0.100\, m_{\mathrm{FCF}} + 2.444\, m_{\mathrm{gross}}, \qquad n = 115,\ R^{2} = 0.46. \tag{6}$$

Gross margin is the dominant explanatory variable; each ten points of gross margin is worth roughly 28% on the multiple, and each ten points of growth about 7%. The fitted multiple gives an implied price; the residual is the mispricing signal that enters the value pillar. Figure 3 shows the universe against the fitted line at median margins.

5.4 Blended fair value

Fair value is 50% DCF, 30% regression and 20% consensus mean target. A method whose output lies outside 0.1–5× the share price is excluded and the weights are renormalised; Nebius loses its DCF this way (negative value under -361% free-cash-flow margins). Table 8 reports the result.

Three readings follow. First, the DCF is negative for eleven of fifteen names; at a 10–11% discount rate the market is paying for growth well beyond the consensus horizon almost everywhere in the chain. Second, the regression says the enablers and memory names trade at one-third to one-half of the multiple their growth and margins would command; that gap—not the DCF—is what puts UCTT, KLIC, CLS, DIOD and DELL on the list. Third, for ALAB, STX and AAOI the blended upside is negative: they are on the list for growth, quality or revisions, and the one-pagers in the accompanying deck treat them as momentum positions with explicit exit thresholds.



Source: SEC EDGAR, Yahoo Finance, FRED; analysis as of 2026-09-02

Figure 3: EV/NTM sales against consensus NTM revenue growth; picks highlighted

Table 8: Valuation by method and market-implied growth

Ticker	Price	Fair value	Upside	DCF	Regression	Target	g^{impl}	EV/S vs fair	P/E NTM
SNDK	$1,537	$2,839	+85%	+17%	+229%	+38%	0%	4.4x / 14.6x	6.9x
MU	$944	$1,580	+67%	−13%	+206%	+60%	11%	4.3x / 13.5x	6.1x
NVDA	$226	$284	+26%	+10%	+39%	+44%	26%	9.5x / 13.3x	17.5x
CRDO	$165	$162	−2%	−38%	+12%	+68%	46%	10.0x / 11.2x	22.2x
ALAB	$271	$180	−34%	−72%	−22%	+44%	$\geq$ 80%	17.4x / 13.5x	48.2x
NBIS	$200	$404	+102%	excl.	+141%	+43%	$\geq$ 80%	6.0x / 14.3x	n/m
KLIC	$79	$98	+24%	−34%	+114%	+35%	75%	2.5x / 5.6x	13.7x
CLS	$282	$376	+33%	−37%	+125%	+69%	48%	1.2x / 2.6x	16.8x
UCTT	$66	$126	+91%	+45%	+156%	+109%	$\geq$ 80%	1.0x / 2.3x	13.2x
AVGO	$369	$416	+13%	−1%	+16%	+43%	31%	11.1x / 12.8x	20.1x
AAOI	$101	$75	−25%	−88%	+20%	+62%	$\geq$ 80%	3.9x / 4.7x	30.4x
APP	$317	$339	+7%	−27%	+27%	+62%	31%	11.1x / 14.1x	17.0x
STX	$797	$524	−34%	−60%	−42%	+41%	50%	9.2x / 5.4x	20.3x
DIOD	$88	$112	+26%	−23%	+71%	+84%	23%	1.7x / 3.1x	17.7x
DELL	$457	$490	+7%	−36%	+70%	+22%	20%	1.5x / 2.5x	18.8x

Source: Pipeline output. "EV/S vs fair" is the actual NTM multiple against the regression-fitted multiple. g^{impl} at the 80% bound means the reverse DCF cannot reach the price with margins held at today's level (UCTT and AAOI have negative current FCF margins).

6 Portfolio construction

6.1 Returns and covariance

Daily total returns over the three years to September 2, 2026 (753 observations) feed a Ledoit–Wolf shrinkage covariance. Three names listed inside the window—SNDK (362 missing days), NBIS (284) and ALAB (136)—are back-filled with the average return of their value-chain segment before listing so that the covariance is complete; this is a proxy, not history, and it understates the idiosyncratic risk of those names. Annualised volatilities range from 47% (KLIC) to 129% (AAOI); the average pairwise correlation is 0.43.

6.2 Black–Litterman expected returns

The prior is CAPM: $\pi_i = r_f + \beta_i \,\mathrm{ERP}$ with realised betas against SPY and a 4.5% premium, which yields 12–16% for the selection. Views are absolute: the blended fair-value upside annualised over a two-year horizon, clipped to $[-20\%, +40\%]$, with view variance $\Omega_{ii} = \tau\Sigma_{ii}/c_i$ where the confidence $c_i \in [0.2, 1]$ rises with the number of agreeing valuation methods and falls with their dispersion. The posterior

$$\mu = \left[(\tau\Sigma)^{-1} + \Omega^{-1}\right]^{-1}\left[(\tau\Sigma)^{-1}\pi + \Omega^{-1}q\right], \qquad \tau = 0.05, \tag{7}$$

ranges from 1.6% (AAOI, negative view) to 20.1% (UCTT). Because $\tau\Sigma$ is large for volatile names, the posterior stays close to the prior unless the view is confident; that is the intended behaviour.

6.3 Optimisation

With h the weight vector, $\widehat{\Sigma}$ the shrinkage covariance and A the segment-membership matrix, the primary portfolio solves

$$\begin{aligned} &\max_{h} \ \frac{h'\mu - r_f}{\sqrt{h'\widehat{\Sigma}h}} \\ &\text{s.t. } \mathbf{1}'h = 1, \qquad 0.02 \le h_i \le 0.15, \qquad Ah \le 0.35, \end{aligned} \tag{8}$$

by sequential quadratic programming from two starting points. Minimum-variance, risk-parity and equal-weight portfolios are reported as references (Table 9); the long-only efficient frontier is drawn without the weight and segment caps (Figure 4).

Table 9: Candidate portfolios

Portfolio	Expected return	Volatility	Sharpe
Maximum Sharpe (recommended)	14.2%	46.5%	0.20
Minimum variance	10.3%	39.0%	0.14
Risk parity	11.1%	44.1%	0.14
Equal weight	11.0%	47.8%	0.13

Source: Black–Litterman expected returns, Ledoit–Wolf covariance, $r_f = 4.75\%$.

The solution is a corner solution in the classical sense: the optimiser buys the names with confident positive views and moderate volatility (UCTT, MU, DIOD, KLIC, NVDA, SNDK) and holds everything with a negative or weak view at the 2% floor. Two consequences matter for implementation. Memory and the two equipment sub-suppliers account for 58% of portfolio variance

Table 10: Weights, risk contributions and return inputs

Ticker	Max Sharpe	Min var	Risk parity	Risk share	μ (BL)	CAPM prior	View	σ
UCTT	15.0%	2.0%	5.8%	18.1%	20.1%	16.0%	38.3%	69%
MU	15.0%	2.0%	5.9%	17.5%	16.8%	15.3%	29.4%	64%
NVDA	13.4%	15.0%	9.1%	8.9%	11.7%	14.0%	12.1%	48%
DIOD	12.5%	13.6%	8.6%	10.1%	13.1%	13.3%	12.5%	55%
KLIC	11.1%	15.0%	9.0%	8.4%	12.5%	12.5%	11.4%	47%
SNDK	11.0%	2.0%	5.4%	14.4%	17.8%	15.6%	35.9%	81%
NBIS	6.0%	2.0%	5.0%	7.3%	17.0%	16.0%	42.2%	92%
CRDO	2.0%	2.0%	4.8%	2.4%	7.7%	16.0%	−1.0%	87%
ALAB	2.0%	2.0%	4.8%	2.3%	3.7%	16.0%	−15.2%	90%
CLS	2.0%	2.0%	5.7%	2.0%	11.5%	15.7%	15.4%	68%
AVGO	2.0%	10.4%	8.0%	1.4%	9.6%	13.8%	6.3%	51%
AAOI	2.0%	2.0%	3.7%	3.1%	1.6%	16.0%	−13.7%	129%
APP	2.0%	7.2%	7.7%	1.2%	8.2%	14.9%	3.4%	76%
STX	2.0%	15.0%	8.5%	1.5%	4.8%	11.8%	−15.2%	53%
DELL	2.0%	7.8%	8.0%	1.4%	8.9%	13.2%	3.5%	61%
Total	100%	100%	100%	100%				

Source: Pipeline outputs `portfolio_weights.csv`, `risk_contributions.csv`, `asset_stats.csv`. Risk share is the max-Sharpe portfolio's variance decomposition. Group weights: silicon 62%, design & equipment 26%, cloud & software 8%, infrastructure 4%.

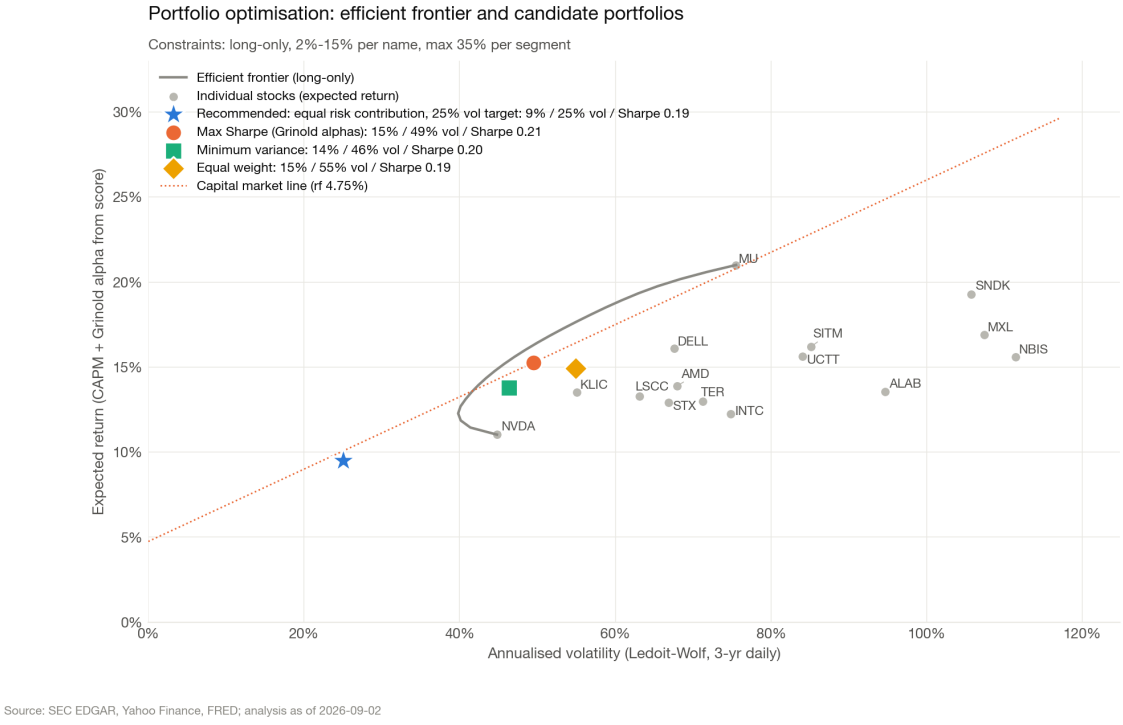


Figure 4: Efficient frontier, individual securities and the four candidate portfolios

against 52% of weight, so the portfolio is a levered bet on the capital-spending cycle even after the segment cap. And the 15% cap binds on three names, which means the recommended weights are as much a statement about the constraints as about the inputs; a mandate with a 10% cap would spread roughly 15 points into NBIS, CLS and AVGO.

6.4 Illustrative backtest

Table 11 and Figure 5 show a monthly-rebalanced constant-mix backtest of the max-Sharpe weights over the same three-year window. It is in-sample by construction: the names were selected and the weights computed with information available only at the end of the window, so the 101% annualised return is a selection artefact and must not be quoted as performance. The exercise is retained because the risk statistics are informative: realised beta of 2.2 to the S&P 500, 47% volatility, a 41% drawdown that coincided with the April 2025 semiconductor sell-off, and 19% tracking error to SOXX.

Table 11: Backtest statistics, September 2023 to September 2026

	CAGR	Volatility	Sharpe	Max drawdown	Total return	Beta vs SPY
Max-Sharpe portfolio (in-sample)	100.7%	46.9%	2.04	−40.7%	+697%	2.24
SOXX	44.9%	39.5%	1.02	−41.4%	+202%	—
QQQ	24.6%	20.5%	0.97	−22.8%	+92%	—
SPY	21.3%	15.3%	1.08	−18.8%	+78%	—

Source: Daily adjusted closes; monthly rebalancing; no transaction costs. Portfolio returns before each name's listing use segment-peer proxies.

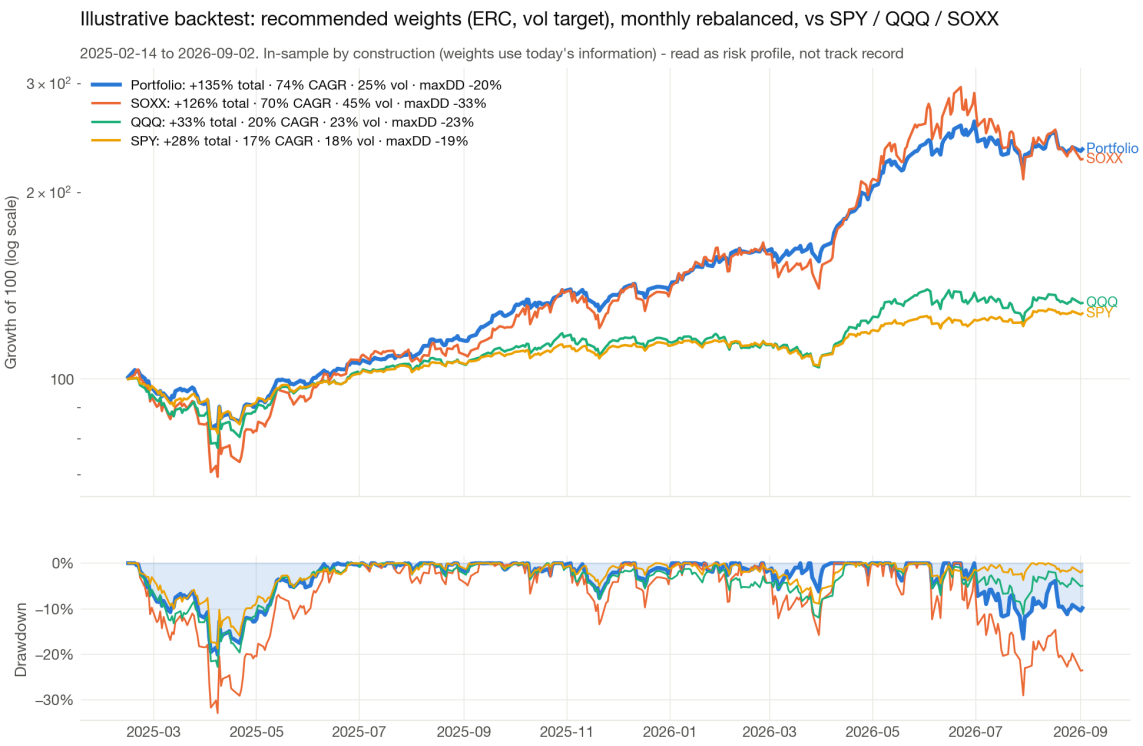
7 Company-level research priorities

7.1 Memory supercycle

SNDK ranks first on every pillar: consensus revenue growth of 120% over the next twelve months, a 79% operating margin, a 6.9× forward P/E and estimates revised up 33% in ninety days. The price has risen 29-fold in a year and the reverse DCF says it discounts zero post-consensus growth; the research question is simply whether NAND pricing holds through the fiscal 2027 supply additions. **MU** is the HBM incumbent with 86% consensus growth into fiscal 2027 and a 6.1× forward P/E; the DCF's through-cycle margin of 15% (against 29% today) is the deliberate haircut for a business whose five-year average free-cash-flow margin is negative. **STX** is the nearline-HDD duopolist; it enters on quality and revisions (+40%) but at 9.2× sales against a regression-fair 5.4× both valuation methods are negative, so it is a momentum position.

7.2 Compute and interconnect

NVDA combines 75% growth with a 47% free-cash-flow margin and 79% ROIC; it is the only large-cap name with a positive DCF (+10%) and its implied growth of 26% is below consensus. Monitoring priorities are custom-silicon substitution, the mix shift to inference, and China. **AVGO** adds custom-ASIC and Ethernet exposure with a 43% FCF margin; upside is modest (+13%) because the DCF is at fair value and the regression residual is small. **CRDO** (active electrical cables, SerDes), **ALAB** (PCIe/CXL retimers and switches) and **AAOI** (800G transceivers) are



Source: SEC EDGAR, Yahoo Finance, FRED; analysis as of 2026-09-02

Figure 5: Growth of 100 and drawdowns, max-Sharpe weights versus benchmarks (in-sample)

the chain's interconnect bottleneck; consensus growth is 76–147% but their prices discount 46% or more post-consensus growth at current margins. Credo fell 20% on the last session of the run and triggered the only critical alert. These three are held at the floor and re-underwritten monthly.

7.3 Manufacturing and infrastructure enablers

UCTT and **KLIC** are the optimiser's largest and fifth-largest positions. Ultra Clean sells gas-delivery subsystems to the equipment OEMs at 1.0× sales with consensus growth of 39%, a 47% revision and a Street target 109% above the price; the model's caveat is a thin 5% operating margin and 3.7× net debt to EBITDA. Kulicke & Soffa's thermo-compression bonders address advanced packaging; it trades at 2.5× sales against a fair 5.6×. **CLS** manufactures hyperscale networking and servers at 12% gross margin—a pass-through business whose 69% growth is worth 2.6× sales on the regression against 1.2× actual. **DELL** and **DIOD** entered the list on the latest run after the valuation recalibration: Dell for AI-server growth and revisions (+34%) at 1.5× sales, Diodes as the cheapest analog name with rising estimates.

7.4 AI monetisation and capacity

NBIS is the highest-growth and lowest-quality name in the selection: 347% consensus growth, capex several times revenue, and only two valuation methods. Its +102% blended upside rests on the regression and the Street; the optimiser gives it 6% because the view confidence is the lowest in the list (0.40). **APP** is the mirror image: a 66% free-cash-flow margin and 69% ROIC, negative revisions and a 34% decline over twelve months. It is on the list as an undervalued quality compounder and receives the floor weight.

8 Monitoring and validation design

8.1 Daily monitoring

The pipeline runs each weekday morning (launchd, 07:15 local) and rebuilds the full chain: data refresh with cache ageing, features, valuation, scoring, portfolio, charts, the investor deck and an HTML dashboard. Each run is archived under `outputs/history/<date>/`, and a run-over-run comparison produces entries, exits, rank moves, weight changes and turnover. Table 12 lists the alert rules; the September 2 run raised 41 alerts (1 critical, 19 warnings, 21 informational).

Table 12: Alert rules on the daily dashboard

Rule	Threshold	Severity	Rationale
One-day price move	$\lvert r\rvert \geq 5\%$ ($\geq 10\%$)	warning (critical)	Event or estimate shock; re-check thesis
30-day EPS revision	$\lvert\Delta\rvert \geq 5\%$	info / warning	Estimate momentum reversal
Blended upside below floor	$< 10\%$	warning	Thesis check, consider trimming
Blended upside above ceiling	$> 60\%$	info	Verify inputs (consensus, FX) before acting
Earnings inside window	14 days	info	Event risk budget
Weight change vs prior run	≥ 3 pts	info	Rebalance trade
One-year drawdown	$\leq -30\%$	warning	Sentiment or fundamental break
New 10-Q / 8-K	within 7 days	info	Refresh model from filing
Top-15 entry / exit	any	info / warning	Selection turnover

8.2 Validation protocol

A single cross-sectional run cannot establish that the score or the optimiser adds value. Before the pillar weights in equation (2) are tuned away from their documented priors, the following protocol should be completed on the point-in-time history that the daily archive now accumulates.

1. **Sample.** Reconstruct the eligible universe at each historical date including delisted names (Pure Storage and Confluent left the universe during development).
2. **Timing.** Lag statements to filing dates and consensus to publication timestamps; rebalance monthly.
3. **Nested walk-forward.** Fit pillar weights, the Black–Litterman confidence map and the post-consensus growth share inside a rolling window; record only the untouched next period.
4. **Costs.** Deduct spread and impact; the small-cap enablers (UCTT, KLIC, DIOD) are the most expensive to trade.
5. **Benchmarks and metrics.** SOXX, the equal-weight eligible universe and a 12–1 momentum strategy; report CAGR, volatility, Sharpe, drawdown, turnover, rank stability and factor information coefficients by semiconductor-cycle regime.
6. **Stress tests.** Shock hyperscaler capex, HBM pricing, gross margins, export controls and the ten-year yield; recompute fair values and the optimiser under each.

The model should be rejected if its advantage disappears under modestly higher costs, small changes in rebalance date, alternative factor definitions, or removal of the best-performing security.

9 Risks and data-quality controls

Several fields required explicit correction during development and are worth recording. Provider enterprise value for ADRs mixed USD market capitalisation with local-currency debt; a first pass produced a 5,784% DCF upside for TSMC before the FX and share-count logic was added. Yahoo's "current-year" consensus for Micron referred to the fiscal year that had already ended, which shifted NTM revenue by a year until the alignment rule was written. The DCF itself was recalibrated three times: an unconstrained fade produced 300% upsides for memory, and a 14% WACC cap with raw betas produced negative values for the entire universe. Every model field now carries a source, a timestamp and a reconciliation rule, and the parameters in Appendix B are versioned with the code.

10 Conclusion

The AI semiconductor theme is best understood as a chain of scarce capabilities whose valuations no longer move together. The pipeline described here scores 116 companies on growth, valuation, quality and estimate momentum, values them with three methods that deliberately disagree, and selects fifteen names that span memory, compute, interconnect, manufacturing enablers and AI capacity. Sandisk, Micron and NVIDIA lead the ranking; the optimiser, which weighs valuation

Table 13: Research risk register

Risk	Transmission mechanism	Monitor
AI capex digestion	Hyperscalers slow accelerator, memory, optics and server orders after a front-loaded build-out; correlations of 0.43 mean the whole list de-rates together	Capex guidance, backlog, lead times, semiconductor IP and PPI series
Memory cycle	HBM/DRAM/NAND pricing reverses as supply additions land; memory is 28% of the max-Sharpe weight	Contract pricing, inventories, wafer starts, revision breadth
Customer concentration	A handful of hyperscalers and NVIDIA drive optics, packaging and EMS demand	Design wins, 10-K concentration disclosures
Export controls and geopolitics	Equipment and accelerator restrictions; Taiwan concentration in leading-edge capacity	Licence rules, geographic revenue
Rates and duration	A 4.75% ten-year keeps discount rates at the 11% cap; DCFs are negative for most of the list	DGS10, implied growth vs consensus
Execution and governance	Thin-margin enablers miss ramps; neocloud financing; auditor or accounting events	Cash conversion, accruals, 8-K filings
Data and model risk	Provider fields mix currencies and fiscal calendars; consensus can be sparse; back-filled returns understate IPO risk	FX and NTM checks, method-dispersion flag, alert ceiling

confidence and covariance rather than rank, concentrates in memory, NVIDIA and the two equipment sub-suppliers whose multiples are furthest below what their fundamentals justify, and holds the priced-for-perfection interconnect names at the floor.

The appropriate next step is not a more aggressive score. It is to let the daily archive accumulate a point-in-time panel, run the validation protocol of Section 8, and only then tune pillar weights, view confidence and the growth fade. Until that work is complete, the ranking and the max-Sharpe allocation in this paper are transparent research priors—useful for organising diligence and for monitoring, but not evidence of an optimal realised strategy.

11 Addendum: validation results and model revision

After the working paper was drafted, the scoring system of Section 4 was tested on a point-in-time reconstruction of the universe from SEC filing dates (105 US filers, 176 month-ends, January 2012 to August 2026; ADRs excluded; consensus fields absent, so the test is a lower bound on the live model). The results changed the model.

11.1 What the documented score predicted

The trailing-only version of the 40/30/20/10 composite had a mean three-month rank IC of -0.022 (Newey–West $t = -1.3$) and -0.083 at twelve months ($t = -3.1$); a permutation test placed it at the 99th percentile of shuffled scores, and the score quintiles were U-shaped (bottom quintile 31% annualised, top 30%, middle 20–21%). The value pillar was the cause: every definition tested—raw multiples, the growth- and margin-adjusted regression residual, a company's own multiple history, and a cheap-and-accelerating interaction—had an IC at or below zero (Table 14). Quality levels carried no information.

Table 14: Pre-registered factor zoo: three-month rank IC, 2012–2026 (selected rows)

Feature	Family	IC 3m	t (HAC)	IC 2012–19	IC 2019–26	IC ex-winners	Verdict
Growth acceleration (quarter YoY − TTM)	growth	+0.033	2.33	+0.040	+0.025	+0.016	passes strict rule
Latest-quarter revenue growth	growth	+0.031	1.72	+0.039	+0.023	+0.013	lenient
12-1 momentum, volatility-scaled	momentum	+0.035	1.62	+0.035	+0.035	+0.017	lenient; 12m IC +0.060
Segment 6-month momentum	momentum	+0.024	1.07	+0.019	+0.029	+0.009	best 1m IC
Accruals (−)	quality	+0.010	0.55	+0.034	−0.015	+0.027	unstable
Gross / operating / FCF margin, ROIC	quality	≈ 0					no information
EV/S residual (growth, margin adjusted)	value	−0.020	−1.10	−0.020	−0.020	+0.004	
Own-history EV/S	value	−0.020	−1.10	−0.002	−0.037	+0.002	
Earnings yield	value	−0.034	−1.56	−0.066	−0.000	−0.034	negative
Size (small − large)	size	−0.023	−0.98	−0.040	−0.006	−0.012	large won

Source: `outputs/validation/factor_research/factor_zoo.csv`. 36 candidates with signs and rationales fixed before evaluation; "ex-winners" drops the ten largest cumulative winners as a survivorship proxy.

11.2 Nested out-of-sample test

Selecting features on the first half of the sample alone (strict rule: $t \geq 2$, positive after excluding winners) returns a single feature, growth acceleration; its second-half IC is +0.025 at three months ($t = 1.3$) and +0.059 at twelve months ($t = 2.1$) over 83 months. A composite of growth acceleration, latest-quarter growth, volatility-scaled momentum and segment momentum has a full-sample three-month IC of +0.039 ($t = 2.3$), positive in both halves (+0.035, +0.043), a top-minus-bottom quintile spread of +17.5% per year and a top-15 CAGR of 36.0% against 26.1% for the equal-weight

universe. Walk-forward weights between the two families, estimated only from ICs known at each date, give an out-of-sample IC of +0.031 ($t = 1.5$).

11.3 Portfolio construction

On the redesigned top 15, fully invested constructions are indistinguishable (Sharpe 1.08–1.13 across equal weight, inverse volatility, minimum variance, equal risk contribution and Grinold max-Sharpe), but a 25% ex-ante volatility target with the remainder in cash raises the Sharpe ratio to 1.17–1.22 and cuts the maximum drawdown from roughly 30% to 20–23%. Equal risk contribution with the volatility target (Sharpe 1.22, drawdown −21%, Calmar 1.37) is adopted as the recommended construction.

11.4 Revised model

The live score is now 50% fundamental momentum (growth acceleration, latest-quarter growth, forward acceleration versus trailing growth, 30- and 90-day EPS revisions), 35% price momentum (volatility-scaled 12-1 momentum, segment momentum, six-month relative strength) and 15% growth-adjusted valuation (regression residual only). Quality is reported as risk flags and no longer ranks. On the same trailing-only reconstruction the revised composite has a three-month IC of +0.026 ($t = 1.6$; permutation $p = 0.013$), a quintile spread of +12.5% per year and an alpha of 8.3% per year against SPY and SOXX ($t = 2.1$); the 15% value tilt is a mandate choice with no evidential support and is the first weight the live archive should re-test. On 2 September 2026 the revised score selects MU, DELL, SNDK, KLIC, MXL, STX, SITM, NBIS, SMCI, UCTT, INTC, COHU, AMD, ALAB and CLS; NVDA, AVGO, CRDO, AAOI, APP and CRWV leave the list. The recommended portfolio holds these at 3–5% each with 46% in cash, because the trailing three-year covariance implies 47% volatility for the fully invested basket; the cash share will fall as realised volatility normalises. Tables 6–10 and the deck reflect the original model and are retained as the documented starting point.

12 Addendum II: the strategic-role pillar (scarcity and risk)

The reference framework reserves a 15% weight for a strategic-role factor scored by an analyst rubric with an evidence log, and warns that its failure modes are narrative bias and stale judgment. The revised model adopts that design with two changes: the judgment is produced by an LLM research process rather than a single analyst, and it is split into the two parts a portfolio manager actually needs—how irreplaceable the company is, and how likely the thesis is to break.

12.1 Rubric and evidence log

Each company is scored 0–1 on eight anchored dimensions (scarcity, lock-in, roadmap, AI exposure, pricing power, and three inverted risk dimensions: competitive threat, geopolitical/regulatory exposure, execution and governance) plus optional customer concentration, with three to eight dated, sourced evidence items, a thesis of at most sixty words, kill criteria and watch items. Assessments are versioned; a new version replaces the old only if it cites evidence dated after the previous

assessment, or at the 90-day scheduled refresh. Two sub-scores enter the decision score:

$$\text{scarcity strength} = 0.40\,\text{scarcity} + 0.20\,\text{lock-in} + 0.15\,\text{roadmap} + 0.15\,\text{AI exp.} + 0.10\,\text{pricing}, \tag{9}$$

$$\text{risk} = 0.7\left[1 - (0.35\,\text{comp} + 0.25\,\text{geo} + 0.20\,\text{exec} + 0.20\,\text{conc})\right] + 0.3\,\text{financial risk}, \tag{10}$$

where financial risk is computed from filings and prices (net debt to EBITDA, negative free cash flow, one-year drawdown and volatility) so that the risk side never rests on the LLM alone. Scarcity strength, shrunk by evidence confidence and set to zero when stale, is the 15% scarcity pillar; the inverted LLM risk is a 25% sub-score of the 10% risk-resilience pillar beside market, financial and valuation resilience. Names with risk ≥ 0.75 are excluded from the list unless scarcity strength is ≥ 0.80, in which case they remain with a 5% weight cap.

12.2 Refresh: moving with the market without narrative drift

Re-assessment is event-driven: a new 10-K, 10-Q or 8-K, a five-day move of 10% or more, a 30-day EPS revision of 10% or more, earnings within seven days, an assessment older than 90 days, or a warning from the real-time event monitor (SEC feed, news, intraday prices) queues the name; up to twelve names are re-researched per daily run with server-side web search, parsed into the schema, and accepted only under the change-control rule. The 90-day news direction decays linearly to zero. Every assessment is archived daily so the pillar's forward IC can be measured exactly as the other pillars are.

12.3 Current state

At the time of writing 116 companies carry evidence-logged assessments, 0 carry the reference paper's Table 6 priors (confidence capped at 0.50) and 0 are queued. Across the assessed names the mean scarcity strength is 0.57 (standard deviation 0.14), the mean combined risk 0.41, and the mean evidence confidence 0.76. Table 15 lists the twelve highest scarcity-strength names and any name removed by the risk gate. The decision list on this basis is MU, DELL, SNDK, STX, KLIC, SITM, MXL, NVDA, NBIS, TER, UCTT, AMD, ALAB, LSCC, INTC.

The pillar is a research prior in the sense of Section 8: it cannot be backtested because no history of assessments exists, and it is labelled as unvalidated in every output until the daily archive supports a forward test. What it adds is precisely the information the filed numbers cannot see—who owns the bottleneck, and what would break the thesis—in a form that is scored, cited, versioned and refreshed on evidence rather than on mood.

A Reproducibility checklist

- Archive raw EDGAR company-facts JSON, Yahoo snapshots and FRED observations with timestamps (`data_cache/`); the pipeline refreshes them on an age threshold rather than overwriting.
- Store the full eligible universe and the selected list at every run (`outputs/history/`).
- Version the universe map, feature directions, winsorisation limits and every valuation and optimiser parameter (`ai_semi_intel/config/`).

Table 15: Strategic-role pillar: highest scarcity strength and risk-gate decisions

Ticker	Segment	Scarcity	LLM risk	Fin. risk	Risk	Pillar	News 90d	Conf.	Gate
NVDA	AI Compute Silicon	0.93	0.45	0.00	0.31	0.87	+0.51	0.85	ok
ASML	Semiconductor Equipment	0.92	0.30	0.02	0.21	0.89	+0.27	0.85	ok
TSM	Foundry, IDM & Packaging	0.91	0.36	0.00	0.25	0.87	+0.45	0.88	ok
AVGO	AI Compute Silicon	0.82	0.43	0.03	0.31	0.79	+0.56	0.78	ok
KLAC	Semiconductor Equipment	0.81	0.39	0.15	0.31	0.78	+0.30	0.85	ok
MU	Memory & Storage	0.79	0.50	0.19	0.40	0.74	+0.79	0.85	ok
GOOGL	Cloud, Hyperscale & Neocl	0.79	0.32	0.00	0.23	0.78	+0.33	0.85	ok
ARM	EDA & Semiconductor IP	0.78	0.48	0.23	0.41	0.73	+0.07	0.78	ok
PLTR	AI Software & Application	0.77	0.42	0.18	0.34	0.74	+0.70	0.80	ok
LRCX	Semiconductor Equipment	0.77	0.46	0.15	0.37	0.73	+0.49	0.85	ok
GEV	Power & Energy for AI	0.77	0.25	0.04	0.19	0.78	+0.24	0.85	ok
LITE	Interconnect, Networking	0.77	0.43	0.26	0.38	0.73	+0.70	0.78	ok
NVTS	Analog, Power & Mixed-Sig	0.43	0.67	0.99	0.77	0.38	+0.15	0.60	excluded
SMCI	Servers & Data Center Inf	0.43	0.81	0.69	0.78	0.38	+0.20	0.70	excluded
SMR	Power & Energy for AI	0.32	0.73	0.92	0.79	0.30	-0.14	0.62	excluded

Source: `outputs/latest/strategic_table.csv`; assessments in `data_cache/strategic/assessments/`. Pillar = 0.75 scarcity strength + 0.25 (1 − risk).

- Recompute every displayed table from the persisted CSVs; the deck and dashboard are generated from the same objects as this paper.
- Save optimiser inputs and outputs: expected returns, prior, views, confidence, covariance, constraints, weights and risk contributions.
- Keep rejected calibrations (unconstrained fade, 14% WACC cap) in the change log so that the final assumptions are not mistaken for the only ones tried.

B Model parameters

Block	Parameter	Value
Screen	Pillar weights (V/G/Q/M)	0.40 / 0.30 / 0.20 / 0.10
	Winsorisation	5th / 95th percentile
	Eligibility	Market cap $\geq$ \$2B; feature coverage $\geq$ 60%; max 3 per segment
Valuation	Risk-free rate	10-year Treasury, 4.75%
	Equity risk premium; tax	4.5%; 15%
	Beta	Blume-adjusted, bounded 0.7–1.4
	WACC bounds	7.5%–11.0%
	Post-consensus growth	60% of FY+1 consensus, cap 30%, fade to 3% by year 8; 10-year horizon
	Margin target	60/40 current vs 5-year average (85/15 if 3-year CAGR $>$ 25%); 5-year convergence
	Blend	50% DCF, 30% regression, 20% Street target; 0.1–5$\times$ sanity filter
Portfolio	Lookback; covariance	3 years daily; Ledoit–Wolf
	Black–Litterman	CAPM prior, $\tau = 0.05$, 2-year view horizon, view clip $[-20\%, +40\%]$
	Constraints	Long-only; 2–15% per name; $\leq$ 35% per segment
	Backtest	Monthly rebalance; SPY, QQQ, SOXX benchmarks

C Selected methodological references

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